RadioCastOS

Quick Start and User Guide - version 1.78.34

RadioCastOS runs your internet radio station from one Windows PC: it plays your music, mixes your microphone, a phone-in caller and other audio into one encode, and sends the result to your streaming server (and, if you set one, a backup server). This guide gets you on air in about 15 minutes, then explains every page of the program.

Part 1 - Quick Start

Before you begin

You need a Windows PC, an internet connection, and the connection details from your streaming host: **server type** (Icecast2 or Shoutcast), **host name**, **port**, **source password** and, if your host gives one, a **mount point** (for example `/live`). Keep your licence key ready if you bought Intermediate or Full.

1. Install

1. Run `RadioCastOS-Setup.exe` and follow the steps. If Windows SmartScreen warns about the publisher, choose **More info**, then **Run anyway** (this does not appear once the installer is code-signed).
2. Start RadioCastOS from the Start menu. ffmpeg, the audio engine, is included - there is nothing else to install.

2. Enter your licence key (Intermediate and Full)

1. Open **Settings & Branding** in the left menu and find **Licence & Package**.
2. Paste your key and press **Activate**. The card shows your package and customer name. No internet is needed.
3. Without a key, RadioCastOS runs as **Basic**, which already includes streaming, mixer, player, library and recording.

3. Connect to your server

1. Open the **Streaming** page. Choose the **Server type**, then fill in **Host**, **Port**, **Source username** and **Password**.
2. Leave **Mount point** blank if your host did not give one; RadioCastOS then sends to `/` and, on Icecast, tries `/stream`.
3. Press **Test Connection**. A green result means the server answered. Then press **Save settings**.

> **Tip:** The source username is usually `source` (Icecast). Your listener or admin password is not the source password.

4. Choose your audio

1. Open the **Audio** page. In the mixer, tick the sources you want to broadcast - your microphone, and a playback or virtual-cable device if you want to capture other programs.
2. Under **Encoding**, pick Codec **mp3**, Bitrate **128** and Sample rate **44100**. This works with every server.
3. With Intermediate or Full, open **System Diagnostics** and press **Run Preflight**, and fix anything shown in red. With Basic, press Test Connection on the Streaming page and watch the level meters while you speak.

5. Load some music

1. Open **Library & Playlist**. Press **Add Folder** and choose your music folder.
2. Select tracks and press **Add Selected** to build a playlist, then press **Play Playlist**.

6. Go live

1. Press **Go Live** at the top of the window (or Ctrl+G). The state changes to ON AIR with a running clock and bitrate. When the server reports it, your listener count appears beside the bitrate.
2. Open your station's own player page or stream URL on another device and confirm that you hear the stream.
3. Press **Stop Stream** in the same place to go off air. If you are on air, you are asked first.

> **Tip:** Use the **On Air** page for day-to-day control. It puts Go Live, Record, AutoDJ, the player, the microphone, talk and caller buttons, breaks and jingles, and Emergency Stop on large buttons.

Part 2 - User Guide

Who is using it: roles

A **role** decides which pages and actions are reachable. The current role shows as a chip in the header. Every start begins at the startup role your installer chose; the role you switch to is never saved, so closing and reopening the program never leaves a presenter signed in as Technician. Moving to a lower role is always free. Moving to a higher role asks for the PIN if one is set.

Role	For	Can open
Operator	Presenters, DJs, church media workers	On Air, Now Playing and the music library.
Producer	Station staff	Everything an Operator can, plus Dashboard, Recording, Scheduler, AI Studio, Intelligence Studio, Presenter Command Center and Analytics.
Technician	The installer or engineer	Every page and setting.

> **Note:** Operators and Producers see friendlier page names: Library & Playlist shows as **Music**, Scheduler & Commercials as **Programs**, AI Studio as **AI Presenter**, Presenter Command Center as **Presenter Desk**, Station Dashboard as **Station** and System Diagnostics as **Health**. The Technician sees the engineering names used in this guide.

The left menu

Pages are grouped in the left menu, and the groups can be collapsed. Press Ctrl+B to show or hide the page list. What you see depends on your package and your role.

Page	What it is for	Group	Package
Dashboard	Live status: on air or off air, time on air, bitrate, reconnects, now playing. Go Live button.	Broadcast	Basic
On Air	The control console: large buttons for everything you do during a show.	Broadcast	Basic
Now Playing	Station name, genre, description, website, and a button to push a song title to listeners.	Broadcast	Basic
Streaming	Server type, host, port, mount point, username, password, Test Connection.	System	Basic
Audio	Mixer sources and encoding (codec, bitrate, sample rate, channels), dead-air alert, capture buffer.	System	Basic
System Diagnostics	Preflight check, Station Health, reconnect, diagnostic report and logs for support.	System	Intermediate
Recording	Save a local copy of your broadcast, in a folder you choose.	Media	Basic
Library & Playlist	Your music library and playlists, with measured loudness.	Media	Basic
Scheduler & Commercials	Clock-driven programming, AutoDJ rotation, commercials, imaging.	Automation	Intermediate
AI Studio	AI presenter: scripts, voices, news reading, Auto Program.	AI Studio	Full
Intelligence Studio, Presenter Command Center, Live Show Copilot	Planning and live-assist tools for the AI presenter and the show in progress.	AI Studio	AI features (Full)
Analytics & Station	Reports about what you broadcast.	Management	Intermediate

Page	What it is for	Group	Package
Station Dashboard, Fleet Console	Several stations and several PCs, remote management and alerts.	Management	Full
Settings & Branding	Your name and logo, licence, operator and reliability options, backups, remote management, alerts.	Administration	Basic (some cards Full)

On Air (control console)

The On Air page shows what is playing and what is next, and the state of the broadcast at a glance. Colours mean: green = running, amber = needs attention, red = stop or danger. A button that cannot be used right now is greyed out and its small caption says why, for example *Needs AutoDJ on*.

Button	What it does
GO LIVE / STOP STREAM	Starts or stops the stream. While on air it shows the time on air; if the server drops, it shows Reconnecting. Stopping a live stream asks first.
RECORD / STOP RECORDING	Saves a local copy to your recording folder. Press again to finish the file.
AUTODJ ON / OFF	Starts or stops 24/7 automation from your playlist and schedule. Turning it off while music plays asks first.
PREVIOUS, PLAY / PAUSE, STOP, NEXT	Transport controls for the player.
MUTE, VOL -, VOL +	Mutes the output and sets its level in steps of 5%. The volume comes back to where it was.
MIC	Mutes or unmutes the microphone on air. The stream keeps running and your Windows microphone level is put back exactly.
MUSIC	Fades the music out on air, or brings it back. Presenter and AI voice are not affected.
TALK	For a live presenter: the microphone opens and the music fades down. Press again and the microphone closes and the music fades back up.
CALLER	Appears when you have ticked **Caller** on a source on the Audio page. Puts the phone-in line on or off air; while it is on, your microphone opens and the music ducks. See `docs\WHATSAPP_CALL_IN.md`.
BREAK NEXT	Arms a commercial break to play after the track that is playing now. It follows your campaign pacing and daily and hourly limits, and is logged as run. Press again (BREAK ARMED) to cancel.
JINGLE NEXT	Arms one jingle (or a sweeper, if you have no jingles) for the next AutoDJ pick. Press again (JINGLE ARMED) to cancel.
STATION ID	Plays your station ID file now. It replaces what is playing, so you are asked first.
EMERGENCY STOP	Stops all audio at once and cancels anything the AI had lined up. You keep manual control.

> **Note:** BREAK NEXT and JINGLE NEXT need AutoDJ on. A break and a jingle cannot both be armed at once: the other button greys out and says so. BREAK NEXT also greys out if no campaign or commercial can play right now.

AI buttons (Full package)

Button	What it does
AUTO PROGRAM	Writes and voices a full program with current news, then puts it on air after a 5-second countdown. Press again to cancel.
PREPARE BREAK	Writes and voices a link for the next break without airing it. Needs the Producer role or higher.
AI ON AIR	Puts the prepared break on air now, or writes and airs one if none is ready. Needs the Producer role or higher.
TAKEOVER	Pauses AutoDJ so you can speak live. Needs AutoDJ and Presenter takeover switched on.
RETURN TO AUTODJ	Ends the presenter or live takeover and hands the station back to AutoDJ.

The program clock

- **NOW** shows the track that is on air, or **Standby** when nothing is playing. It never shows the next item in place of the current one.
- **NEXT** and **FOLLOWING** come from the broadcast queue, or from the playlist when the queue is empty. With AutoDJ on and nothing queued, NEXT says *Chosen by AutoDJ*.
- Remaining time uses one rule everywhere. When a clock is loaded, the next post is named (for example NEWS IN 4:10). With no clock it is labelled **WALL**, so wall-clock time is not mistaken for the traffic log.
- The source line shows the encoder clock and bitrate. If the encoder stalls or the playhead drifts away from it, the line turns red.

Streaming (connection)

Choose **icecast2** for Icecast servers, **shoutcast2** for Shoutcast v2, and **shoutcast1** (legacy) only for old Shoutcast v1 servers.

- RadioCastOS opens the connection itself, like the popular butt encoder: it tries the modern request first, then the older one, and accepts either.
- Mount point: leave it blank, or `/`, if your host has none. A mount typed without its slash gets one.
- A wrong password stops at once with a clear message instead of retrying forever.
- **Keep reconnecting whatever the error** (Connection page, off by default): turn it on for unattended stations so a dropped line or a stuck encoder is retried automatically.
- Use TLS only if your server supports https sources.

Backup stream (second mount)

If you have a second server or mount, enter it under **Settings & Branding > Operator & Reliability > Backup stream (host:port/mount)**. When it is set, a backup encode starts with Go Live and stops with it. The backup is fed from the same capture as the main stream, so both mounts carry exactly the same programme. This also works with the native source setting switched on.

Station Health checks that the address is valid. The header shows a **FALLBACK** chip, and warns once per session, if your fallback audio or backup stream is not ready.

Audio and encoding

Setting	Guidance
Codec	mp3 works everywhere. aac gives better quality at low bitrates. ogg, opus and flac go to Icecast2 only.
Bitrate (kbps)	128 is a good default for music. 64 to 96 for speech or limited bandwidth. 192 to 320 for high quality.
Sample rate	44100 is the standard. Opus always runs at 48000; RadioCastOS resamples and says so in the log.
Channels	Stereo for music. Mono halves the data and suits talk stations.
Dead-air alert	After this many silent seconds (0 = off), the log says so and the monitor raises a Dead air fault. Sound returning clears it. Audio is never changed.
Capture buffer (ms)	0 uses the default. Raise it (for example 200 or 500) if the stream drops out or crackles.

- Opus: bitrate 6 to 510 kbps. FLAC is lossless, has no bitrate setting and uses far more bandwidth than MP3. Check that your listeners and host can handle it.
- Shoutcast servers accept MP3 or AAC only. If you choose Opus or FLAC with Shoutcast, RadioCastOS stops before connecting and tells you what to change. Shoutcast v1 always sends MP3.

Mixer and the encode

Each source (microphone, playback device or virtual cable) has its own fader. Use a virtual audio cable to capture another program, and keep your microphone and music at levels where speech is clearly louder than the background.

Microphone, music and caller are mixed in **one** encode. The music ducks while someone talks, a closed caller line is held at zero, and a **master limiter** (about -1 dBTP) stops peaks from clipping on the way out. You do not need to set any of this up.

Recording

On the Recording page choose a folder with **Browse...**, pick a File format (mp3, aac, ogg, opus or flac), then press **Start Recording**. Set **New file every N minutes** to split long shows into files with no gap; each file is named with its own time stamp. Recording is independent of the live stream, so you can record while off air.

Library and playlists

- **Add Folder** scans a folder for audio files: mp3, wav, flac, ogg, aac, m4a and wma. Rescanning picks up new files.
- **Add Selected** moves tracks to the playlist. Drag tracks to reorder them, or drag them back to the library to remove them. You can also drop files and folders from Windows Explorer.
- **Shuffle**, **Clear**, **Remove Duplicates** and **Missing Files** keep the list tidy. Duplicates (same song in different folders) are skipped automatically.
- **Save** and **Load** keep named playlists. **Import M3U** and **Export M3U** exchange playlists with other programs.
- Ctrl+Z undoes and Ctrl+Y redoes any playlist change, including Clear and Shuffle.
- **Loudness:** the library shows each track's measured loudness (LUFS) and marks tracks that are much louder than the rest as **HOT**. Imported files are measured, and unmeasured tracks are measured in the background. Measurements are kept when you save, reload or refresh tags.
- When an AI break is prepared, a cart more than 3 dB above your loudness target is refused, so it cannot jump out of the mix.

Now Playing

Fill in your station name, genre, description and website so listeners' players show them. Song titles are sent automatically as tracks change. To send a title yourself, type it into Now playing and press **Push now-playing update**; the stream does not restart. The program also writes the current title to `nowplaying.json` for use by other tools.

Scheduler, commercials and AutoDJ (Intermediate and Full)

The scheduler runs your station by the clock: music rotation, station imaging (jingles and sweepers), and commercial campaigns at set times. Turn on **AUTODJ** from the On Air page to run it around the clock. Set up the schedule first, then test it for a few minutes while off air.

- Campaign pacing, daily and hourly limits and anti-repeat rules apply to every break, including one you arm with BREAK NEXT.
- Every aired spot is written to the **as-run log** and the affidavit, which records the time the encoder actually sent it.
- The Commercial Break Planner and Composer, and Traffic Intelligence, help you build and check breaks before they air.

Station Health and the Emergency Media Vault

On **System Diagnostics**, press **Station Health** for one page that shows the state of the emergency stop, the stream, silence detection, the backup stream, the Emergency Media Vault and open incidents. The older Health, Failover, Continuity and Incident centres open from this page.

- The **Emergency Media Vault** checks that the fallback and emergency audio you configured really exists, decodes, and is not silent. Use **Choose folder...** to point it at your fallback audio.
- These centres are **advisory**. They tell you what is ready and what is not. They never start playback, change the queue, switch streams or bypass Emergency Stop.
- **Run Preflight**, **Reconnect Stream** and **Export Diagnostic Report** are on the same page; send the report when you ask for help.

AI Studio (Full)

The AI presenter writes and voices short links between tracks. It can work online (you supply your own OpenAI key) or offline (a local writer and the voices installed on Windows, with no key and no internet). It can also read news from the web. **Auto Program** builds a presenter break in one click. You can fall back to offline automatically if the online service fails. Only clone a voice with the written consent of the person it belongs to.

- **Piper** and **Kokoro** are optional offline neural voices. They are not included with RadioCastOS: each runs from a separate install that you provide. Open **PIPER SETUP** or **KOKORO SETUP...** in AI Presenter Studio to download them to your own user folder or point to files you already have. Voice models have their own licences; check them before broadcasting.
- **Presenter Command Center** prepares and controls presenter breaks and Autopilot. **Live Show Copilot** suggests talking points and next steps while the show runs. **Intelligence Studio** is where you set up the intelligence that feeds them.
- If the chosen voice is not available, the presenter falls back to the next one and the log says why.

Several stations and remote control (Full)

- Station profiles keep separate settings for each station on one PC (passwords are never written into profile files). Activate a profile and restart the program to load it.
- **Remote Management** (Settings & Branding) is off by default. When on, it serves a token-protected status page to this PC only, or to your local network if you choose. Remote Go Live and Stop is a separate tick box, also off by default.
- **Fleet Console** watches other stations from one screen; **Alerts** warn you about lasting stream or system problems, and can send them to a webhook. See `docs\REMOTE-MANAGEMENT.md` for details.

Packages

	Basic	Intermediate	Full
Streaming, mixer, player, library, recording, dashboard, On Air	Yes	Yes	Yes
Scheduler, AutoDJ, commercials, analytics, diagnostics, alerts		Yes	Yes
AI presenter and studio, voice cloning, station profiles, fleet console, remote management, white-label branding			Yes

Your key works offline and is checked every time the program starts. When a key expires, the package keeps working for 14 more days, then the program returns to Basic. To renew, paste the new key over the old one in Licence & Package. A key belongs to the customer named on it; please do not share it.

Keyboard shortcuts

Keys	Action
Ctrl+G	Go live / stop stream (asks first when on air)
Space	Play / pause
Alt+Left, Alt+Right	Previous / next track
Ctrl+Left, Ctrl+Right	Seek back / forward 10 seconds
Ctrl+Up, Ctrl+Down	Volume up / down
M	Mute
Ctrl+1 to Ctrl+9	Jump to the pages in the left menu, in order
Ctrl+PgUp, Ctrl+PgDn	Previous / next page
Ctrl+B	Show or hide the page list
Ctrl+L	Show or hide the connection log
Enter, Delete	In the playlist: add or play the selected track, remove it from the playlist
Alt+Up, Alt+Down	Move the selected playlist track up / down
Ctrl+Z, Ctrl+Y	Undo / redo playlist changes
F1	Show all shortcuts

Troubleshooting

Problem	What to try
Test Connection fails or Go Live will not connect	Check host, port and password with your streaming host. Make sure the server type matches (Icecast2 or Shoutcast). Try leaving Mount point blank. Open the log with Ctrl+L: it explains the failure in plain language.
"Wrong password" message	Use the source password from your host, not the admin or listener password. Check the source username (often source).
Stream connects but drops out	Raise Capture buffer on the Audio page, lower the bitrate, turn on Keep reconnecting whatever the error, and check your upload speed (you need more than your bitrate).
No audio device found	Plug in the microphone, then restart RadioCastOS (with Intermediate or Full, also press Run Preflight on System Diagnostics). Check that Windows privacy settings allow desktop apps to use the microphone.
Listeners hear silence	Check the mixer faders and the MIC and MUSIC buttons. Turn on the Dead-air alert (for example 30 seconds) so you are told.
BREAK NEXT or JINGLE NEXT is greyed out	Read the small caption on the button. Usually AutoDJ is off, no commercials or jingles are loaded, or the other one is already armed.
The source line on the desk is red	The encoder has stalled or the player has drifted away from it. Open Station Health. Stopping and restarting the stream clears the warning once the cause is gone.
The header shows FALLBACK	Your fallback audio or backup stream is not ready. Open Station Health and Emergency Media Vault to see which check failed.
A page is missing from the left menu	It may need a higher role or package. Check the role chip in the header and your package in Licence & Package.
Opus or FLAC refused	The server is Shoutcast. Choose mp3 or aac, or use an Icecast2 server.
Licence key says not valid	Paste the whole key again with no extra spaces. If it still fails, send us the customer name on the key. Keys cannot be edited. If the message mentions the computer clock, correct the Windows date.

Your files and getting help

RadioCastOS keeps your settings in `%APPDATA%\RadioCastOS\config.json` and its logs in the logs folder beside it. If a settings, library or schedule file cannot be read, the program backs it up beside the original, keeps the newest three backups, and tells you at start-up. Back up your settings before moving to a new PC, or use the deployment export in Settings & Branding (it leaves out passwords and your media).

When you ask for help, send the newest log file, the Diagnostic Report from System Diagnostics, and the exact message you saw. Contact CircuitBridge Technologies, Benin City, Nigeria, through circuitbridgetechnologies.name.ng or the details on your purchase confirmation.

Streaming services have their own rules. You are responsible for holding the music and broadcast licences your station needs.